Jovan Vasić

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EDUCATION

Technical University of Munich (TUM) **Oct. 2026 – Present**
M.Sc. Mechatronics, Robotics and Biomechanical Engineering *Munich, Germany*

- Coursework: Robotics, Advanced Control, Autonomous Systems, Introduction to Deep Learning

University of Novi Sad, Faculty of Technical Sciences **2021 – 2026**
B.Eng. Mechatronics Engineering, Major in Robotics and Automation *Novi Sad, Serbia*

- **GPA: 8.70/10**; Serbian Ministry of Education Scholarship for Gifted Students (2022–2023)
- Thesis: Autonomous robotic cable insertion on a UR5e using intelligent perception and impedance control

WORK EXPERIENCE

RT-RK **May 2025 – Jul. 2025**
Embedded Software Intern, Edge AI *Novi Sad, Serbia*

- Built an ESP32-S3 smart thermostat on FreeRTOS with an LVGL touchscreen interface.
- Trained and benchmarked setpoint-prediction models; deployed a quantized PyTorch MLP on-device via ESP-DL.

DunavNET **Aug. 2024 – Oct. 2024**
Computer Vision Intern *Novi Sad, Serbia*

- Developed object detection and segmentation pipelines in Python and OpenCV.

PROJECTS & COMPETITIONS

Intrinsic AI for Industry Challenge **Mar. 2026 – Aug. 2026**
Team Lead

- Led the team to **4th of 160+ teams in the qualification round**; 3/3 successful insertions on both connector types.
- Built a full autonomy pipeline for cable-connector insertion on a UR5e.
  - **Stack:** ROS 2; Gazebo; YOLOv8; SVD/DLT triangulation; impedance control; F/T sensing; GCP
- Transferred the system to a real xArm7 cell; paper in preparation for *IEEE Robotics and Automation Practice*.

Eurobot Robotics Team **Sep. 2023 – Present**
Software Lead *Novi Sad, Serbia*

- Placed **5th in Europe** (Eurobot 2024) and 5th in Serbia (Eurobot 2026).
- Wrote the STM32H7 control stack from scratch in C: 1 kHz PID/odometry, motion profiling, multi-FSM strategy.
  - **Stack:** C; STM32 HAL; Raspberry Pi 5; ROS 2; OpenCV; ArUco; LiDAR
- Published a paper on YOLO detection trained on synthetic data rendered in Unreal Engine 5.

Autonomous Quadcopter **2026**
Personal Project *C++ / ROS 2 / Gazebo*

- Implemented the mission state machine and position controller for an autonomous trash-collecting drone.

SKILLS & INTERESTS

- **Robotics:** ROS 2; Gazebo; impedance control; force/torque sensing; PID; motion planning
- **Perception & ML:** OpenCV; Deep Learning; multi-view geometry; PyTorch; SciPy; model quantization
- **Embedded & Software:** C; C++; Python; STM32; ESP32; FreeRTOS; Linux; Git
- **Languages:** Serbian (native); English (fluent); German (A1); French (basic)
- **Interests:** Hobby electronics; combat sports; outdoor activities